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Q1) European call option price = \$4.20
Stock price = 38 (S)
Strike price k = 35
$T = 0.33$ yrs
rate = 6%

$$C = SN(d_1) - ke^{-rT}N(d_2)$$

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$\sigma \in [0.1, 0.5]$ Tried

$\sigma = 0.2$

$$d_1 = \frac{\ln(38/35) + (0.06 + (0.2)^2/2)(0.333)}{0.2\sqrt{0.333}}$$

$\simeq 0.0830$

$d_2 = 0.714$

$C \simeq 4.13$

$\sigma = 0.3$

$d_1 = 0.565$ $d_2 = 0.3917$

$C = 4.80$

$\sigma = 0.25$

$d_1 = 0.670$ $d_2 = 0.526$

$C = 4.43$

$\sigma = 0.23$

$C = 4.31$

$\sigma = 0.20 \Rightarrow c = 4.13$

$\sigma = 0.23 \Rightarrow c = 4.31$

$$\sigma = 0.20 + \frac{4.20 - 4.13}{4.31 - 4.13}(0.23 - 0.20)$$

≈ 0.2117

Implied volatility ≈ 0.212

(b) Put call parity

$$P = C + Ke^{-rT} - S$$

$$P = 4.20 + 35\, e^{-0.06 \times 0.33} - 38$$

$\simeq 0.507$

Put price $\approx \$0.51$

(c) PV = \$38M

cost to launch (exercise price) = \$35 M

(a) $\sigma \approx 0.28$ $C = 4.20$

option value = \$4.20M

NPV if launch = \$3M

option worth more

so waiting is better.

Question 2 → | Shelly Singhal

Current stock price $S_0 = 100$
Strike price $K = 105$
Time to Maturity $T = 10$ days
Option type: European call

Part A) Discrete Binomial Model

Daily :- \$1 or - \$1 $p = 0.5$

After 10 days: $S_T = 100 + (2k - 10)$ k = no. of up moves

$S_T = 2k + 90$

a) Prob that option ends in the money.

$S_T > 105$ $2k + 90 > 105$ $k > 7.5$ $\boxed{k \geq 8}$

$\therefore k = 8, 9, 10$

$P(\text{in the money}) = \sum_{k=8}^{10} \binom{10}{k} \left(\frac{1}{2}\right)^{10}$

$= \frac{1}{1024} \left({}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}\right) = \frac{1}{1024}(45 + 10 + 1)$

$= \frac{56}{1024} \approx 0.0547$

b) Expected Payoff

$k = 8$ $S_T = 106 \Rightarrow$ Payoff $= 1$
$k = 9$ $S_T = 108 \Rightarrow$ Payoff $= 3$
$k = 10$ $S_T = 110 \Rightarrow$ Payoff $= 5$

$E[\text{Payoff}] = \sum_{k=8}^{10} \text{Payoff}_k \cdot P(k)$

$= \frac{1}{1024}(1 \cdot 45 + 3 \cdot 10 + 5 \cdot 1)$

$= \frac{1}{1024}(45 + 30 + 5) = \frac{80}{1024} = 0.0781$

c) Fair value = expected payoff = \$0.0781

Part B) Continuous Normal Distribution Model

a) Find σ

Given, $E[|x|] = 1$

$$E[|x|] = \sigma\sqrt{\frac{2}{\pi}} \Rightarrow \sigma = \frac{1}{\sqrt{2/\pi}} = \sqrt{\frac{\pi}{2}} = 1.2533$$

Over 10 days, the standard deviation of cumulative return

$$\sigma_{10} = \sigma\sqrt{10} \approx 1.2533 \cdot \sqrt{10} \approx 3.962$$

b) Expected Payoff

$$S_T \sim N(100, \sigma_{10}^2 = 15.7)$$

$$E[\max(S_T - 105, 0)] = \int_{105}^{\infty} (S-105) f_{S_T}(S)\,dS$$

$$Z = \frac{S-100}{3.962} \qquad S \Rightarrow 100 + 3.962Z$$

$$E[\max(S_T - 105, 0)] = \int_{z_0}^{\infty} (100 + 3.962z - 105) \cdot \phi(z)\,dz$$

$$= \int_{z_0}^{\infty} (-5 + 3.962z)\phi(z)\,dz$$

lower bound

$$z_0 = \frac{105-100}{3.962} \approx 1.262 \qquad \phi(z) \to \text{standard normal}$$

$$E[\max(S-K, 0)] = (S_0 - K)\Phi(d) + \sigma_T \phi(d)$$

$$d = \frac{S_0 - K}{\sigma_T} = \frac{-5}{3.962} \approx -1.262$$

$$E[\text{payoff}] \approx \sigma_T \phi(d) - (K - S_0)\Phi(d)$$

$$\phi(-1.262) \approx 0.1812$$

$$\Phi(-1.262) \approx 0.1035$$

$$= 3.962 \times 0.1812 - 5 \times 0.1035$$

$$\approx 0.7177 - 0.5175 = 0.2002$$

c) Numerical Evaluation $\rightarrow$ \$0.2002

a) support $[a, b]$ $E[|x|] = 1$

expected absolute move still \$1

$X \sim U[a, b]$ $E[|x|] = 1$

If symmetric, $X \sim U[-c, c]$

$$E[|x|] = \int_{-c}^{c} \frac{|x|}{2c}\,dx = \frac{1}{c}\int_0^c x\,dx = \frac{1}{c}\frac{c^2}{2} = \frac{c}{2}$$

$$\frac{c}{2} = 1 \qquad \boxed{c = 2}$$

so support is $[-2, 2]$

b) $\mathrm{Var}(X) = \frac{(2-(-2))^2}{12} = \frac{16}{12} = \frac{4}{3}$, mean $= 0$

$$\sigma_{10} = \sqrt{10 \cdot \frac{4}{3}} \approx 3.651$$

c) simulation Method:

We can use Monte-carlo simulation

Define the model: $E[|\Delta S|] = 1$

$\Delta S \sim U[-2, 2]$

$E[|\Delta S|] = \frac{a}{2} = 1$

$a = 2$

$\Delta S \sim U[-2, 2]$

simulation procedure

$S_0 = 100$, $K = 105$, $T = 10$ days.

No. of simulations = 100000 (for ep)

Each simulation: generate 10 daily price change

$\Delta S_1, \Delta S_2 \ldots \Delta S_{10}$ each from $U[-2, 2]$

Terminal stock price $S_T = S_0 + \sum_{i=1}^{10} \Delta S_i$

Payoff $= \max(S_T - K, 0)$

Estimate Fair Value

$$= \frac{1}{N}\sum_{i=1}^{N} \max(S_T^{(i)} - K, 0)$$

Assignment - 4

Subject

Date: / /

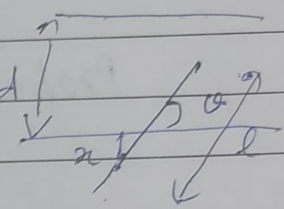
MON TUE WED THR FRI SAT SUN

(3) Buffon's Needle- Monte Carlo experiment of π

P(needle intersect one of the lines) = ?

$$f(x,\theta) = \frac{4}{\pi d} \qquad 0 \le x \le d/2$$

$$0 \le \theta \le \pi/2$$



Intersect if $x \le \frac{l}{2}\sin\theta$

$$P\left(x \le \frac{l}{2}\sin\theta\right) = \int_0^{\pi/2}\int_0^{l/2\,\sin\theta} \frac{4}{\pi d}\,dx\,d\theta$$

$$= \int_0^{\pi/2} \frac{4}{\pi d}\cdot\frac{l}{2}\sin\theta\,d\theta$$

$$= \int_0^{\pi/2} \frac{2l}{\pi d}\sin\theta\,d\theta$$

$$= \int_0^{\pi/2} \frac{2l}{d\pi}\sin\theta\,d\theta$$

$$= \frac{2l}{\pi d}\left[-\cos\pi/2 + \cos 0\right]$$

$$P\left(x \le \frac{l}{2}\sin\theta\right) = \frac{2l}{\pi d}$$

$$\boxed{\pi \simeq \frac{2l}{d \times (\text{No. of crossing})}}$$

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